Array | Submit | 0 | 01:18:23 | Premium

Description | Editorial | Solutions | Submissions | Run Ctrl

35. Search Insert Position

Solved

Easy | Topics | Companies

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [1,3,5,6], target = 5
Output: 2
```

Example 2:

```
Input: nums = [1,3,5,6], target = 2
Output: 1
```

Example 3:

```
Input: nums = [1,3,5,6], target = 7
Output: 4
```

Constraints:

18.6K | 425 | 253 Online

Code

Java | Auto

```java
class Solution {
    public int searchInsert(int[] nums, int target) {

        int low = 0, high = nums.length - 1;

        while (low <= high) {

            int mid = low + (high - low) / 2;

            if (nums[mid] == target)
                return mid;

            else if (nums[mid] < target)
                low = mid + 1;

            else
                high = mid - 1;
```

Saved | Ln 1, Col 1

Testcase | Test Result

Output

2

Expected

2

Array | Submit | 0 | 01:19:44 | Premium

Description | Editorial | Solutions | Submissions

39. Combination Sum

Medium | Topics | Companies

Given an array of **distinct** integers `candidates` and a target integer `target`, return *a list of all* ***unique combinations*** *of* `candidates` *where the chosen numbers sum to* `target`. You may return the combinations in **any order**.

The **same** number may be chosen from `candidates` an **unlimited number of times**. Two combinations are unique if the frequency of at least one of the chosen numbers is different.

The test cases are generated such that the number of unique combinations that sum up to `target` is less than `150` combinations for the given input.

Example 1:

```
Input: candidates = [2,3,6,7], target = 7
Output: [[2,2,3],[7]]
Explanation:
2 and 3 are candidates, and 2 + 2 + 3 = 7. Note that 2 can be used multiple
times.
7 is a candidate, and 7 = 7.
These are the only two combinations.
```

Example 2:

20.8K | 223 | 264 Online

Code

Java | Auto

```java
            res.add(new ArrayList<>(temp));
            return;
        }

        if (target < 0 || idx == a.length)
            return;

        // pick current element
        temp.add(a[idx]);
        backtrack(idx, a, target - a[idx], temp, res);
        temp.remove(temp.size() - 1);

        // skip current element
        backtrack(idx + 1, a, target, temp, res);
    }
}
```

Saved | Ln 29, Col 2

Testcase | Test Result

[2,3,6,7]

target =

7

Output

[[2,2,3],[7]]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

40. Combination Sum II

Medium | Topics | Companies

Given a collection of candidate numbers (`candidates`) and a target number (`target`), find all unique combinations in `candidates` where the candidate numbers sum to `target`.

Each number in `candidates` may only be used **once** in the combination.

Note: The solution set must not contain duplicate combinations.

Example 1:

```
Input: candidates = [10,1,2,7,6,1,5], target = 8
Output:
[
[1,1,6],
[1,2,5],
[1,7],
[2,6]
]
```

Example 2:

```
Input: candidates = [2,5,2,1,2], target = 5
Output:
```

12.2K | 225 | 134 Online

Code

Java | Auto

```
        return;
    }

    for (int i = start; i < a.length; i++) {

        // skip duplicates
        if (i > start && a[i] == a[i - 1]) continue;

        if (a[i] > target) break;
```

Saved | Ln 31, Col 2

Testcase | Test Result

Case 1 | Case 2

Input

candidates =
[10,1,2,7,6,1,5]

target =
8

Output

[[1,1,6],[1,2,5],[1,7],[2,6]]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

45. Jump Game II

Medium | Topics | Companies

You are given a **0-indexed** array of integers `nums` of length `n`. You are initially positioned at index 0.

Each element `nums[i]` represents the maximum length of a forward jump from index `i`. In other words, if you are at index `i`, you can jump to any index `(i + j)` where:

- `0 <= j <= nums[i]` and
- `i + j < n`

Return *the minimum number of jumps to reach index* `n - 1`. The test cases are generated such that you can reach index `n - 1`.

Example 1:

```
Input: nums = [2,3,1,1,4]
Output: 2
Explanation: The minimum number of jumps to reach the last index is 2. Jump 1
step from index 0 to 1, then 3 steps to the last index.
```

Example 2:

```
Input: nums = [2,3,0,1,4]
Output: 2
```

16.4K | 273 | 185 Online

Code

Java | Auto

```
        if (i == end) {
            jumps++;
            end = far;
        }
    }
    return jumps;
}
}
```

Saved | Ln 17, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

nums =
[2,3,1,1,4]

Output

2

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

49. Group Anagrams

Medium | Topics | Companies

Given an array of strings `strs`, group the anagrams together. You can return the answer in **any order**.

Example 1:

```
Input: strs = ["eat","tea","tan","ate","nat","bat"]
Output: [["bat"],["nat","tan"],["ate","eat","tea"]]
Explanation:
- There is no string in strs that can be rearranged to form "bat".
- The strings "nat" and "tan" are anagrams as they can be rearranged to form each other.
- The strings "ate", "eat", and "tea" are anagrams as they can be rearranged to form each other.
```

Example 2:

```
Input: strs = [""]
Output: [[""]]
```

Example 3:

21.9K | 382 | 398 Online

Code

Java | Auto

```
        String key = new String(ch);

        map.putIfAbsent(key, new ArrayList<>());
        map.get(key).add(s);
    }

    return new ArrayList<>(map.values());
}
}
```

Saved | Ln 18, Col 2

Testcase | Test Result

strs =
["eat","tea","tan","ate","nat","bat"]

Output

[["eat","tea","ate"],["bat"],["tan","nat"]]

Expected

[["bat"],["nat","tan"],["ate","eat","tea"]]

Contribute a testcase

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

66. Plus One

Easy | Topics | Companies

You are given a **large integer** represented as an integer array `digits`, where each `digits[i]` is the ith digit of the integer. The digits are ordered from most significant to least significant in left-to-right order. The large integer does not contain any leading `0`'s.

Increment the large integer by one and return *the resulting array of digits.*

Example 1:

```
Input: digits = [1,2,3]
Output: [1,2,4]
Explanation: The array represents the integer 123.
Incrementing by one gives 123 + 1 = 124.
Thus, the result should be [1,2,4].
```

Example 2:

```
Input: digits = [4,3,2,1]
Output: [4,3,2,2]
Explanation: The array represents the integer 4321.
Incrementing by one gives 4321 + 1 = 4322.
Thus, the result should be [4,3,2,2].
```

11.7K | 672 | 180 Online

Code

Java | Auto

```java
        digits[i] = 0;
    }

    // if all digits were 9
    int[] res = new int[n + 1];
    res[0] = 1;
    return res;
  }
}
```

Saved | Ln 18, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

digits =

[1,2,3]

Output

[1,2,4]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

73. Set Matrix Zeroes

Medium | Topics | Companies | Hint

Given an `m x n` integer matrix `matrix`, if an element is `0`, set its entire row and column to `0`'s.

You must do it in place.

Example 1:

1	1	1
1	0	1
1	1	1

⇒

1	0	1
0	0	0
1	0	1

```
Input: matrix = [[1,1,1],[1,0,1],[1,1,1]]
Output: [[1,0,1],[0,0,0],[1,0,1]]
```

Example 2:

0	1	2	0		0	0	0	0

17K | 324 | 183 Online

Code

Java | Auto

```java
            for (int j = 0; j < n; j++)
                matrix[0][j] = 0;

        // first column
        if (firstCol)
            for (int i = 0; i < m; i++)
                matrix[i][0] = 0;
    }
}
```

Saved | Ln 42, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

matrix =

[[1,1,1],[1,0,1],[1,1,1]]

Output

[[1,0,1],[0,0,0],[1,0,1]]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

74. Search a 2D Matrix

Medium | Topics | Companies

You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `true` *if* `target` *is in* `matrix` *or* `false` *otherwise.*

You must write a solution in `O(log(m * n))` time complexity.

Example 1:

1	3	5	7
10	11	16	20
23	30	34	60

Code

Java | Auto

```java
class Solution {
    public boolean searchMatrix(int[][] matrix, int target) {
        int m = matrix.length;
        int n = matrix[0].length;

        int low = 0, high = m * n - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;
            int row = mid / n;
```

Saved | Ln 1, Col 1

Testcase | Test Result

Input

matrix =

[[1,3,5,7],[10,11,16,20],[23,30,34,60]]

target =

3

Output

true

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

75. Sort Colors

Medium | Topics | Companies | Hint

Given an array `nums` with `n` objects colored red, white, or blue, sort them **in-place** so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers `0`, `1`, and `2` to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Example 1:

```
Input: nums = [2,0,2,1,1,0]
Output: [0,0,1,1,2,2]
```

Example 2:

```
Input: nums = [2,0,1]
Output: [0,1,2]
```

Constraints:

- `n == nums.length`

21.4K | 650 | 202 Online

Code

Java | Auto

```
        }
    }

    void swap(int[] nums, int i, int j) {
        int temp = nums[i];
        nums[i] = nums[j];
        nums[j] = temp;
    }
}
```

Saved | Ln 21, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

nums =
[2,0,2,1,1,0]

Output

[0,0,1,1,2,2]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

78. Subsets

Medium | Topics | Companies

Given an integer array `nums` of **unique** elements, return *all possible subsets (the power set)*.

The solution set **must not** contain duplicate subsets. Return the solution in **any order**.

Example 1:

```
Input: nums = [1,2,3]
Output: [[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]
```

Example 2:

```
Input: nums = [0]
Output: [[],[0]]
```

Constraints:

- `1 <= nums.length <= 10`
- `-10 <= nums[i] <= 10`
- All the numbers of `nums` are **unique**.

19.1K | 238 | 209 Online

Code

Java | Auto

```
        for (int i = 0; i < size; i++) {
            List<Integer> temp = new ArrayList<>(res.get(i));
            temp.add(num);
            res.add(temp);
        }
    }
    return res;
}
}
```

Saved | Ln 18, Col 2

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

nums =
[1,2,3]

Output

[[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

79. Word Search

Medium | Topics | Companies

Given an `m x n` grid of characters `board` and a string `word`, return `true` *if* `word` *exists in the grid.*

The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

Example 1:

A	B	C	E
S	F	C	S
A	D	E	E

```
Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCCED"
Output: true
```

17.6K | 334 | 192 Online

Code

Java | Auto

```
        boolean found = dfs(board, word, i + 1, j, idx + 1) ||
                        dfs(board, word, i - 1, j, idx + 1) ||
                        dfs(board, word, i, j + 1, idx + 1) ||
                        dfs(board, word, i, j - 1, idx + 1);

        board[i][j] = temp; // backtrack
        return found;
    }
}
```

Saved | Ln 33, Col 2

Testcase | Test Result

Input

board =
[["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]]

word =
"ABCCED"

Output

true

Expected

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

18. 4Sum

Medium | Topics | Companies

Given an array `nums` of `n` integers, return *an array of all the* ***unique*** *quadruplets* `[nums[a], nums[b], nums[c], nums[d]]` such that:

- `0 <= a, b, c, d < n`
- `a`, `b`, `c`, and `d` are **distinct**.
- `nums[a] + nums[b] + nums[c] + nums[d] == target`

You may return the answer in **any order**.

Example 1:

```
Input: nums = [1,0,-1,0,-2,2], target = 0
Output: [[-2,-1,1,2],[-2,0,0,2],[-1,0,0,1]]
```

Example 2:

```
Input: nums = [2,2,2,2,2], target = 8
Output: [[2,2,2,2]]
```

Code

Java | Auto

```java
                if (sum == target) {
                    res.add(Arrays.asList(nums[i], nums[j], nums[l], nums[r]));
                    l++; r--;

                    while (l < r && nums[l] == nums[l - 1]) l++;
                    while (l < r && nums[r] == nums[r + 1]) r--;
                }
                else if (sum < target) {
                    l++;
```

Saved | Ln 39, Col 1

Testcase | Test Result

Accepted Runtime: 2 ms

Case 1 | Case 2

Input

nums =
[1,0,-1,0,-2,2]

target =
0

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

33. Search in Rotated Sorted Array

Medium | Topics | Companies

There is an integer array `nums` sorted in ascending order (with **distinct** values).

Prior to being passed to your function, `nums` is **possibly left rotated** at an unknown index `k` (`1 <= k < nums.length`) such that the resulting array is `[nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]]` (**0-indexed**). For example, `[0,1,2,4,5,6,7]` might be left rotated by `3` indices and become `[4,5,6,7,0,1,2]`.

Given the array `nums` **after** the possible rotation and an integer `target`, return *the index of* `target` *if it is in* `nums`*, or* `-1` *if it is not in* `nums`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [4,5,6,7,0,1,2], target = 0
Output: 4
```

Example 2:

```
Input: nums = [4,5,6,7,0,1,2], target = 3
Output: -1
```

Example 3:

29.8K | 559 | 308 Online

Code

Java | Auto

```java
            if (nums[mid] == target)
                return mid;

            // left half sorted
            if (nums[low] <= nums[mid]) {
                if (target >= nums[low] && target < nums[mid])
                    high = mid - 1;
                else
                    low = mid + 1;
            }
            // right half sorted
```

Saved | Ln 28, Col 2

Testcase | Test Result

Case 1 | Case 2 | Case 3

Input

nums =
[4,5,6,7,0,1,2]

target =
0

Output

4

Array | Submit | 0 | 01:19:59 | Premium

Description | Editorial | Solutions | Submissions

34. Find First and Last Position of Element in Sorted Array

Medium | Topics | Companies

Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given `target` value.

If `target` is not found in the array, return `[-1, -1]`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [5,7,7,8,8,10], target = 8
Output: [3,4]
```

Example 2:

```
Input: nums = [5,7,7,8,8,10], target = 6
Output: [-1,-1]
```

Example 3:

```
Input: nums = [], target = 0
Output: [-1,-1]
```

23.1K | 354 | 190 Online

Code

Java | Auto

```java
            if (nums[mid] == target) {
                ans = mid;
                high = mid - 1;
            } else if (nums[mid] < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        return ans;
```

Saved | Ln 43, Col 2

Testcase | Test Result

Case 1 | Case 2 | Case 3

Input

nums =
[5,7,7,8,8,10]

target =
8

Output

[3,4]